

Outcomes and content for Stage 5

Financial mathematics A

Outcomes

A student:

- develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly **MAO-WM-01**
- solves financial problems involving simple interest, earning money and spending money **MA5-FIN-C-01**

Related Life Skills outcomes: MALS-FIN-01, MALS-FIN-02

Content

Solve problems involving earning money

- Solve problems involving wages given an hourly rate of pay including penalty rates for overtime, weekends and public holidays
- Calculate earnings from non-wage sources exploring [commission](#), piece work and royalties
- Calculate weekly, fortnightly, monthly and yearly earnings assuming 1 year = 52 weeks
- Calculate leave loading by finding a percentage of eligible normal pay
- Investigate sources of published tables or online calculators and use these to calculate the weekly, fortnightly or monthly tax to be deducted from a worker's pay under the Australian Pay-As-You-Go (PAYG) taxation system
- Determine annual [taxable income](#) by exploring allowable deductions and current tax rates
- Calculate net earnings after deductions and taxation

Solve problems involving simple interest

- Establish and use the formula $I = Prn$ to find simple interest where I = simple interest, P = principal, r = interest rate per time period and n = number of time periods
- Apply the [simple interest](#) formula to solve problems related to investing money at simple interest rates, both algebraically and graphically

Solve problems involving spending money

- Calculate the cost of buying items on terms, by paying an initial deposit and making regular repayments
- Examine payment options involving *buy now, pay later* and investigate the costs associated with these schemes for purchasing goods
- Examine the principles behind short-term loans involving small dollar amounts and compare borrowing costs associated with using these products

Financial mathematics B

Outcomes

A student:

- develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly **MAO-WM-01**
- solves financial problems involving compound interest and depreciation **MA5-FIN-C-02**

Related Life Skills outcomes: MALS-FIN-01, MALS-FIN-02

Content

Solve problems involving compound interest and depreciation

- Examine [compound interest](#) for up to 3 time periods using repetition of the formula for [simple interest](#)
- Associate the calculation of the total value of a compound interest investment with repeated multiplication, using digital tools
- Establish and use the formula $FV = PV(1 + r)^n$ to find compound interest where FV = future value of the investment, PV = present value of the investment, r = interest rate per time period and n = number of time periods
- Solve problems involving compound interest
- Compare simple interest with compound interest in practical situations
- Use the compound interest formula to establish the depreciation formula $S = V_0(1 - r)^n$ where S = salvage value, V_0 = initial value of the asset, r = depreciation rate per time period and n = number of periods
- Solve problems involving the [depreciation](#) of an asset

Algebraic techniques A

Outcomes

A student:

- develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly **MAO-WM-01**
- simplifies algebraic fractions with numerical denominators and expands algebraic expressions **MA5-ALG-C-01**

Related Life Skills outcomes: MALS-PAT-01

Content

Apply the 4 operations to simplify algebraic fractions with numerical denominators

- Simplify expressions that involve [algebraic fractions](#) with numerical [denominators](#)

Apply the distributive law to the expansion of algebraic expressions, and collect like terms where appropriate

- Expand [algebraic expressions](#), including those with negative [coefficients](#)
- Expand and simplify algebraic expressions by removing grouping symbols and collecting like terms where appropriate
- Expand [binomial](#) products algebraically using the [distributive law](#) or consider areas of [rectangles](#) as a possible method of expansion

Indices A

Outcomes

A student:

- develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly **MAO-WM-01**
- simplifies algebraic expressions involving positive-integer and zero indices, and establishes the meaning of negative indices for numerical bases **MA5-IND-C-01**

Content

Extend and apply the index laws to variables, using positive-integer indices and the zero index

- Apply the [index](#) laws for numerical [bases](#) with positive-[integer indices](#) to develop the index laws in algebraic form
- Establish that $x^0 = 1$ algebraically using index laws
- Simplify [algebraic expressions](#) that involve the zero index

Simplify algebraic products and quotients using index laws

- Simplify algebraic expressions that involve powers, products and [quotients](#) of simple [algebraic terms](#) containing positive-integer indices

Apply index laws to numerical expressions with negative-integer indices

- Apply index notation, [patterns](#) and index laws to establish the meaning of negative indices for numerical bases
- Evaluate numerical expressions involving a negative index by first representing them with a positive index
- Represent given numbers in index form (integer indices and bases only) and vice versa

Equations A

Outcomes

A student:

- develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly **MAO-WM-01**
- solves linear equations of up to 3 steps, limited to one algebraic fraction **MA5-EQU-C-01**

Related Life Skills outcomes: MALS-ADS-01, MALS-MDI-01

Content

Solve linear equations involving up to 3 steps

- Solve [linear equations](#) using algebraic techniques involving up to 3 steps
- Solve linear equations with pronumerals on both sides of the equation
- Solve linear equations involving grouping symbols
- Verify solutions using substitution

Solve linear equations involving one algebraic fraction

- Solve linear equations involving one [algebraic fraction](#) using up to 3 steps

Solve linear equations arising from word problems and substitution into formulas

- Solve linear equations arising from substitution into formulas
- Represent word problems as linear equations, solve the equations and interpret the solutions in the context of the problem

Linear relationships A

Outcomes

A student:

- develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly **MAO-WM-01**
- determines the midpoint, gradient and length of an interval, and graphs linear relationships, with and without digital tools **MA5-LIN-C-01**

Related Life Skills outcomes: MALS-POS-01

Content

Find the midpoint and gradient of a line segment (interval) on the Cartesian plane

- Plot and join 2 points to form an [interval](#) on the [Cartesian plane](#) and use the interval as the [hypotenuse](#) of a right-angled [triangle](#)
- Apply the relationship gradient $m = \frac{\text{rise}}{\text{run}}$ to find the gradient/slope of the interval joining the 2 points
- Distinguish between intervals with positive and negative [gradients](#) from a diagram
- Explain why horizontal intervals have a gradient of 0 and vertical intervals have undefined gradients using the gradient relationship
- Determine the [midpoint](#) of horizontal and vertical intervals on the Cartesian plane
- Apply the process for calculating the mean to find the midpoint, M of the interval joining 2 points on the Cartesian plane
- Use graphing applications to find the midpoint and gradient/slope of an interval

Find the distance between 2 points located on the Cartesian plane

- Use the interval between 2 points as the hypotenuse of a right-angled triangle on the Cartesian plane and apply [Pythagoras' theorem](#) to determine the [length](#) of the interval joining the 2 points
- Use graphing applications to find the distance between 2 points on the Cartesian plane

Recognise and graph equations

- Recognise that equations of the form $y = mx + c$ represent linear relationships or straight lines
- Construct tables of values and use [coordinates](#) to graph a variety of [linear relationships](#) on the Cartesian plane, with and without digital tools
- Identify the x- and y- intercepts of lines
- Determine whether a point lies on a [line](#) using substitution

Examine parallel, horizontal and vertical lines

- Explain that [parallel](#) lines have equal gradients/slopes
- Explain why the x-axis has the equation $y = 0$ and the y-axis has the equation $x = 0$
- Recognise $y = c$ as a line parallel to the x-axis and $x = k$ as a line parallel to the y-axis
- Graph vertical and horizontal lines

Linear relationships B

Outcomes

A student:

- develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly **MAO-WM-01**
- graphs and interprets linear relationships using the gradient/slope-intercept form **MA5-LIN-C-02**

Related Life Skills outcomes: MALS-POS-01

Content

Examine the gradient/slope-intercept form

- Interpret the coefficient of $x(m)$ as the gradient/slope, and the constant (c) as the y -intercept for equations of the form $y = mx + c$
- Find the equation of a straight line in the form $y = mx + c$, given the gradient/slope and the y -intercept of the line
- Graph equations of the form $y = mx + c$ by using the gradient and the y -intercept
- Determine the gradient and y -intercept of a straight line from its graph and apply these values to determine the equation of the line
- Explain the effect of increasing or decreasing the [gradient](#) with or without digital tools
- Recognise and describe [linear relationships](#) in real-life contexts

Find the equations of parallel and perpendicular lines

- Explain that 2 straight lines are perpendicular if the product of their gradients is -1
- Find the equation of a straight line that is parallel or perpendicular to another given line by applying $y = mx + c$

Non-linear relationships A

Outcomes

A student:

- develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly **MAO-WM-01**
- identifies connections between algebraic and graphical representations of quadratic and exponential relationships in various contexts **MA5-NLI-C-01**

Content

Examine the connection between algebraic and graphical representations of quadratics and exponentials

- Represent quadratic and [exponential relationships](#) using graphing applications
- Construct a table of values to [graph non-linear](#) relationships involving quadratics and exponentials
- Explain that quadratic relationships are represented by [parabolas](#)
- Identify graphs and [equations](#) of parabolas and exponential [curves](#)
- Recognise quadratics and exponentials in real-life contexts

Non-linear relationships B

Outcomes

A student:

- develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly **MAO-WM-01**
- identifies and compares features of parabolas and exponential curves in various contexts **MA5-NLI-C-02**

Content

Graph and examine quadratic relationships

- Graph and compare parabolas of the form $y = kx^2$ and $y = kx^2 + c$ using graphing applications
- Identify and describe the key features of parabolas of the form $y = kx^2$ and $y = kx^2 + c$ including the vertex, x- and y-intercepts, axis of symmetry and concavity

Graph and examine exponential relationships

- Graph and compare exponential curves of the form $y = a^x$ using graphing applications
- Describe features of exponential curves including the y-intercept, asymptote and the nature of the curve for very large and very small values of x

Distinguish between linear, quadratic and exponential relationships by examining their graphical representations

- Associate [graphs](#) of straight [lines](#), [parabolas](#) and exponential [curves](#) with the appropriate [equations](#)
- Recognise [non-linear](#) relationships in real-life contexts and solve related problems
- Use graphing applications to solve a pair of [simultaneous equations](#) where one equation is non-linear and interpret the solution

Numbers of any magnitude

Outcomes

A student:

- develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly **MAO-WM-01**
- solves measurement problems by using scientific notation to represent numbers and rounding to a given number of significant figures **MA5-MAG-C-01**

Content

Identify and describe very small and very large measurements

- Identify and describe the meaning of common prefixes, such as *milli*, *centi* and *kilo*
- Establish the meaning of prefixes for very small or very large measurement units

Find absolute and percentage error

- Determine the precision of a measuring instrument by finding the smallest division on the instrument
- Find the absolute error of measuring instruments ($absolute\ error = \frac{1}{2} \times precision$)
- Calculate the percentage error of a given measurement by applying the formula:

$$percentage\ error = \frac{absolute\ error}{measurement} \times 100\%$$

Estimate and round numbers to a specified degree of accuracy

- Apply the language of estimation appropriately, including the terms [rounding](#), [approximate](#) and *level of accuracy*
- Round numbers to a specified number of [significant figures](#)
- Examine the effect that truncating or rounding during calculations has on the [accuracy](#) of the results

Express numbers in scientific notation

- Recognise the need for notation to express very large or very small numbers
- Represent numbers using [scientific notation](#) in practical contexts
- Order numbers expressed in scientific notation
- Represent numbers expressed in scientific notation as a [decimal](#)
- Estimate the value of calculations involving scientific notation by applying knowledge of [index laws](#)
- Solve problems with calculations involving scientific notation using digital tools

Trigonometry A

Outcomes

A student:

- develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly **MAO-WM-01**
- applies trigonometric ratios to solve right-angled triangle problems **MA5-TRG-C-01**

Content

Demonstrate and explain the constancy of trigonometric ratios for a given angle in right-angled triangles

- Identify and label the [hypotenuse](#), [adjacent](#) and opposite sides with respect to a given [angle](#) in a right-angled [triangle](#) in any orientation
- Define the sine, cosine and tangent ratios for angles in right-angled triangles and use trigonometric notation $\sin \theta$, $\cos \theta$, $\tan \theta$
- Identify the [sine](#), [cosine](#) and [tangent ratios](#) in a right-angled triangle
- Verify the constancy of the sine, cosine and tangent ratios for a given angle by applying knowledge of similar right-angled triangles
- Find approximations of the [trigonometric ratios](#) for a given angle
- Find the size of an angle given one of the trigonometric ratios for the angle using digital tools

Apply trigonometry to solve right-angled triangle problems

- Apply trigonometry to find the [lengths](#) of unknown [sides](#) in right-angled triangles with a given angle including angles in [degrees](#) and minutes
- Apply trigonometry to find the size of unknown angles in right-angled triangles including in degrees and minutes
- Solve a variety of practical problems involving trigonometric ratios in right-angled triangles

Trigonometry B

Outcomes

A student:

- develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly **MAO-WM-01**
- applies trigonometry to solve problems, including bearings and angles of elevation and depression **MA5-TRG-C-02**

Content

Solve right-angled triangle problems involving angles of elevation and depression

- Identify and describe [angles of elevation](#) and [depression](#)
- Solve practical problems involving angles of elevation and depression

Solve right-angled triangle problems involving bearings

- Identify and interpret [true bearings](#) and [compass bearings](#)
- Explain the difference between true bearings and compass bearings and convert between them
- Solve practical problems involving bearings

Area and surface area A

Outcomes

A student:

- develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly **MAO-WM-01**
- solves problems involving the surface area of right prisms and practical problems involving the area of composite shapes and solids **MA5-ARE-C-01**

Related Life Skills outcomes: MALS-ARE-01

Content

Solve problems involving areas and surface areas

- Solve practical problems involving the areas of [composite shapes](#)
- Identify the edge [lengths](#) and the faces making up the [surface area](#) of prisms
- Recognise and justify whether a diagram represents a [net](#) of a right prism
- Create and rearrange nets of right prisms
- Find the surface areas of prisms, given their nets, excluding [curved surfaces](#)
- Solve problems involving surface areas of prisms, excluding curved surfaces

Develop and apply the formula for surface areas of cylinders

- Recognise the curved surface of a [cylinder](#) as a [rectangle](#) and apply this knowledge to calculate the area of the curved surface
- Develop and apply the formula to find the surface area of a closed cylinder: $A = 2\pi r^2 + 2\pi rh$, where r is the length of the radius and h is the perpendicular height

Solve problems involving surface areas of cylinders and related composite solids

- Solve problems involving surface areas of cylinders and related composite solids

Volume A

Outcomes

A student:

- develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly **MAO-WM-01**
- solves problems involving the volume of composite solids consisting of right prisms and cylinders **MA5-VOL-C-01**

Related Life Skills outcomes: MALS-VOL-01

Content

Solve problems involving composite solids consisting of right prisms and cylinders

- Find the [volumes](#) of composite right prisms with [uniform cross-sections](#) that may be dissected into [triangles](#) and quadrilaterals
- Find the volumes of right prisms that have uniform cross-sections in the form of [sectors](#), semicircles and quadrants
- Calculate volumes of composite solids consisting of right prisms and [cylinders](#)
- Solve practical problems related to the volumes and [capacities](#) of composite solids

Properties of geometrical figures A

Outcomes

A student:

- develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly **MAO-WM-01**
- identifies and applies the properties of similar figures and scale drawings to solve problems **MA5-GEO-C-01**

Related Life Skills outcomes: MALS-GEO-01

Content

Identify and describe the properties of similar figures

- Describe similar figures as having the same [shape](#) but not necessarily the same size
- Verify and explain that in similar [polygons](#), the [corresponding angles](#) are equal and the corresponding [side lengths](#) are in the same proportion
- Name the vertices in matching order when using the similar symbol ($\sim$) in a similarity statement
- Match the corresponding sides and [angles](#) of similar polygons

Solve problems using ratio and scale factors in similar figures

- Apply an appropriate scale to enlarge or reduce a diagram
- Determine the scale factor for pairs of similar polygons and [circles](#)
- Apply knowledge of scale factor to find unknown sides in similar polygons
- Solve problems involving unknown lengths and scale factors of similar figures and related practical problems
- Solve problems involving scale drawings, with or without digital tools

Data analysis A

Outcomes

A student:

- develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly **MAO-WM-01**
- compares and analyses datasets using summary statistics and graphical representations **MA5-DAT-C-01**

Related Life Skills outcomes: MALS-DAT-02

Content

Examine standard deviation as a measure of spread

- Identify [standard deviation](#) as a [measure of spread](#)
- Calculate the [standard deviation](#) of a small [dataset](#) using digital tools
- Compare small datasets using standard deviation

Determine quartiles and interquartile range

- Determine the 5-number summary for [sets](#) of numerical [data](#)
- Determine the 5-number summary from [graphical](#) representations
- Determine the [interquartile range \(IQR\)](#) for datasets
- Compare and explain the relative merits of [range](#) and IQR as measures of spread

Represent datasets using box plots and use them to compare datasets

- Represent numerical datasets using a [box plot](#) to display the [median](#), upper and lower [quartiles](#), and maximum and minimum values
- Compare 2 or more numerical datasets using [parallel](#) box plots drawn on the same scale
- Compare and contrast the centres, spreads and shapes of 2 or more numerical datasets, using box plots and numerical statistics, including the 5-number summary
- Determine quartiles from datasets displayed in histograms and dot plots, and represent these as a box plot
- Identify and describe [skewness](#) or [symmetry](#) of datasets displayed in histograms, dot plots and box plots
- Interpret box plots to draw conclusions and make [inferences](#) about the dataset

Data analysis B

Outcomes

A student:

- develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly **MAO-WM-01**
- displays and interprets datasets involving bivariate data **MA5-DAT-C-02**

Related Life Skills outcomes: MALS-DAT-02

Content

Identify and describe numerical datasets involving 2 variables

- Distinguish between situations involving 1-[variable](#) and 2-variable [\(bivariate\) data](#) and explain when each is needed
- Explain the difference between variables that have an association and variables that have a [causal](#) relationship
- Identify and describe the [independent variable](#) and [dependent variable](#) in relationships with possible cause and effect

Represent datasets involving 2 numerical variables, using a scatter plot and a line of best fit, by eye

- Gather [data](#) on a topic of interest involving 2 [numerical variables](#)
- Represent the data using a [scatter plot](#)
- Create a [line of best fit](#), by eye, on an existing scatter plot

Interpret data involving 2 numerical variables, using graphical representations

- Describe informally the association between 2 numerical variables and apply terminology about form (linear), strength (strong, moderate or weak) and direction (positive or negative)
- Use the line of best fit, by eye, to make predictions between known data values ([interpolation](#)) and what might happen beyond known data values (extrapolation)
- Explain the limitations of the model when making predictions

Probability A

Outcomes

A student:

- develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly **MAO-WM-01**
- solves problems involving probabilities in multistage chance experiments and simulations **MA5-PRO-C-01**

Related Life Skills outcomes: MALS-PRO-01

Content

Describe multistage chance experiments involving independent and dependent events

- Explain the difference between [dependent](#) and [independent events](#) in experiments involving 2 stages
- Explain how the [probability](#) of independent and dependent events differs in relation to replacement

Solve problems for multistage chance experiments

- Record all possible [outcomes](#) for multistage [chance](#) experiments
- Determine the probabilities of outcomes for multistage independent events using $P(A \text{ and } B) = P(A) \times P(B)$, where necessary
- Determine the probabilities of outcomes for multistage dependent events
- Associate [complementary events](#) with probabilities in multistage chance experiments

Design and use simulations to model and examine situations involving probability

- Design and conduct a probability [simulation](#), modelling probabilities of events, using digital tools
- Record and use the results of a probability simulation to predict future events
- Apply reasoning to evaluate the simulation and its related outcomes

Variation and rates of change A (Path)

Outcomes

A student:

- develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly **MAO-WM-01**
- identifies and solves problems involving direct and inverse variation and their graphical representations (*Path: Stn, Adv*) **MA5-RAT-P-01**

Content

Stn (Standard), Adv (Advanced) and Ext (Extension) have been used to suggest paths for related Stage 6 courses.

Identify and describe problems involving direct and inverse variation

- Describe typical examples of [direct variation/proportion](#)
- Apply the language of direct variation to everyday contexts: y is directly proportional to x , y is proportional to x , y varies directly as x
- Identify and represent direct variation/proportion as $y \propto x$ (y is proportional to x) or $y = kx$, where k is the constant of variation
- Describe typical examples of [inverse \(indirect\) variation](#)
- Apply the language of inverse variation to everyday contexts: y is inversely proportional to x , y is proportional to the reciprocal of x , y varies inversely as x
- Identify and represent inverse variation/proportion as $y \propto \frac{1}{x}$ (y is inversely proportional to x) or $y = \frac{k}{x}$, where k is the constant of variation

Identify and describe graphs involving direct and inverse variation

- Recognise and describe direct variation from graphs, noting that the graph of $y = kx$ is a straight line passing through the origin, with its gradient k being the constant of variation
- Recognise and describe inverse variation from graphs, noting that the graph of $y = \frac{k}{x}$ is a curve

Solve problems involving direct and inverse variation and examine the relationship between graphs and equations corresponding to proportionality

- Solve problems involving direct or inverse variation using an [equation](#)
- Use linear conversion [graphs](#) to convert from one unit to another
- Graph equations representing direct variation, with or without digital tools

Variation and rates of change B (Path)

Outcomes

A student:

- develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly **MAO-WM-01**
- analyses and constructs graphs relating to rates of change (*Path: Stn, Adv*) **MA5-RAT-P-02**

Content

Stn (Standard), Adv (Advanced) and Ext (Extension) have been used to suggest paths for related Stage 6 courses.

Analyse graphs that are decreasing or increasing at a constant rate

- Describe [direct variation graphs](#) as increasing at a [constant](#) rate and interpret the [gradient](#) of the [line](#) as the rate of change
- Interpret and analyse further graphs that exhibit a constant rate of change, including those that are decreasing at a constant rate

Analyse the relationship between graphs and variable rates of change

- Interpret distance–time graphs when the speed is [variable](#), describing the rate of increase or decrease, the initial and final points, constant relationships represented by straight lines and variable relationships represented by [curved](#) lines
- Describe the rate of change between variables in a variety of contexts including direct and [inverse variation](#)
- Describe qualitatively the rate of change of a graph, using terms such as *increasing at a decreasing rate*

Construct graphical representations of rates of change

- Represent a given description of a variable rate of change of one quantity over time

Algebraic techniques B (Path)

Outcomes

A student:

- develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly **MAO-WM-01**
- simplifies algebraic fractions involving indices, and expands and factorises algebraic expressions (*Path: Adv*) **MA5-ALG-P-01**

Content

Stn (Standard), Adv (Advanced) and Ext (Extension) have been used to suggest paths for related Stage 6 courses.

Apply the 4 operations involving algebraic fractions with pronumerals in the denominator

- Simplify [algebraic fractions](#), including those involving [indices](#)
- Simplify [expressions](#) that involve [operations](#) with algebraic fractions, including algebraic fractions that involve [pronumerals](#) in the [denominator](#) and/or indices

Factorise algebraic expressions by taking out a common algebraic factor

- [Factorise algebraic expressions](#), including those involving indices, by determining the [highest common factors \(HCF\)](#) including negative [coefficients](#)/pronumerals

Expand binomial products and factorise monic quadratic expressions

- Expand [binomial](#) products
- Factorise [monic quadratic](#) trinomial expressions
- Solve problems involving expansion and factorisation of algebraic expressions

Algebraic techniques C (Path)

Outcomes

A student:

- develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly **MAO-WM-01**
- selects and applies appropriate algebraic techniques to operate with algebraic fractions, and expands, factorises and simplifies algebraic expressions (*Path: Adv*) **MA5-ALG-P-02**

Content

Stn (Standard), Adv (Advanced) and Ext (Extension) have been used to suggest paths for related Stage 6 courses.

Operate with algebraic fractions involving binomial numerators and numerical denominators

- Add and subtract [algebraic fractions](#) with [binomial numerators](#) and numerical [denominators](#)

Expand, factorise and simplify algebraic expressions including special products

- Prove and apply these special products $(a - b)(a + b) = a^2 - b^2$, $(a + b)^2 = a^2 + 2ab + b^2$,
 $(a - b)^2 = a^2 - 2ab + b^2$
- Expand and simplify a variety of [algebraic expressions](#) including binomial products and the special products
- [Factorise](#) algebraic expressions involving the special products and strategies, including [common factors](#), grouping in pairs for 4-term expressions and quadratic trinomials ([monic](#) and non-monic)
- Simplify algebraic expressions involving algebraic fractions using factorisation

Indices B (Path)

Outcomes

A student:

- develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly **MAO-WM-01**
- applies the index laws to operate with algebraic expressions involving negative-integer indices (Path: Adv) **MA5-IND-P-01**

Content

Stn (Standard), Adv (Advanced) and Ext (Extension) have been used to suggest paths for related Stage 6 courses.

Apply index laws to algebraic expressions involving negative-integer indices

- Apply index notation, patterns and index laws to establish $a^{-1} = \frac{1}{a}$, $a^{-2} = \frac{1}{a^2}$, $a^{-3} = \frac{1}{a^3}$ and

$$a^{-n} = \frac{1}{a^n}$$

- Represent [expressions](#) involving negative-[integer indices](#) as expressions involving positive-integer indices and vice versa
- Apply the [index](#) laws to simplify algebraic products and [quotients](#) involving negative-integer indices
- Describe and use x^{-1} as the reciprocal of x and generalise this relationship to expressions of the form $\left(\frac{a}{b}\right)^{-1}$
- Use knowledge of the reciprocal to simplify expressions of the form $\left(\frac{a}{b}\right)^{-n}$

Indices C (Path)

Outcomes

A student:

- develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly **MAO-WM-01**
- describes and performs operations with surds and fractional indices (*Path: Adv*) **MA5-IND-P-02**

Content

Stn (Standard), Adv (Advanced) and Ext (Extension) have been used to suggest paths for related Stage 6 courses.

Describe surds

- Describe a real number as a number that can be represented by a point on the [number line](#)
- Examine the differences between [rational](#) and [irrational numbers](#) and recognise that all rational and irrational numbers are real
- Convert between [recurring decimals](#) and their [fractional](#) form using digital tools
- Describe the term *surd* as referring to irrational expressions of the form $\sqrt[n]{x}$ where x is a rational number and n is an integer such that $n \geq 2$, and $x > 0$ when n is even
- Recognise that a [surd](#) is an exact value that can be [approximated](#) by a [rounded decimal](#)
- Demonstrate that $\sqrt{x}$ is undefined for $x < 0$ and that $\sqrt{x} = 0$ when $x = 0$ using digital tools
- Describe $\sqrt{x}$ as the positive square root of x for $x > 0$ and $\sqrt{0} = 0$

Apply knowledge of surds to solve problems

- Establish and apply the following results for $x > 0$ and $y > 0$: $(\sqrt{x})^2 = x = \sqrt{x^2}$, $\sqrt{xy} = \sqrt{x} \times \sqrt{y}$
and $\sqrt{\frac{x}{y}} = \frac{\sqrt{x}}{\sqrt{y}}$
- Apply the 4 [operations](#) to simplify [expressions](#) involving surds
- Expand and simplify expressions involving surds
- Rationalise the denominators of surds of the form $\frac{a\sqrt{b}}{c\sqrt{d}}$

Describe and use fractional indices

- Apply index laws to describe fractional indices as: $a^{\frac{1}{n}} = \sqrt[n]{a}$ and $a^{\frac{m}{n}} = \sqrt[n]{a^m} = (\sqrt[n]{a})^m$
- Translate expressions in surd form to expressions in [index](#) form and vice versa
- Evaluate numerical expressions involving fractional [indices](#) including using digital tools

Equations B (Path)

Outcomes

A student:

- develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly **MAO-WM-01**
- solves monic quadratic equations, linear inequalities and cubic equations of the form $ax^3 = k$ (*Path: Adv*) **MA5-EQU-P-01**

Content

Stn (Standard), Adv (Advanced) and Ext (Extension) have been used to suggest paths for related Stage 6 courses.

Solve monic quadratic equations

- Solve quadratic equations of the form $ax^2 + bx + c = 0$, limited to $a = 1$, using factors

Solve cubic equations

- Determine that for any value of k there is a unique value of x that solves a cubic equation of the form $ax^3 = k$ where $a \neq 0$
- Solve cubic equations of the form $ax^3 = k$, leaving answers in exact form and as decimal approximations

Solve linear inequalities and graph their solutions on a number line

- Represent [inequalities](#) on a [number line](#)
- Solve linear inequalities, including those with negative numbers, and [graph](#) the solutions
- Recognise that an inequality has an infinite number of solutions unless other restrictions are introduced

Equations C (Path)

Outcomes

A student:

- develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly **MAO-WM-01**
- solves linear equations of more than 3 steps, monic and non-monic quadratic equations, and linear simultaneous equations (*Path: Adv*) **MA5-EQU-P-02**

Content

Stn (Standard), Adv (Advanced) and Ext (Extension) have been used to suggest paths for related Stage 6 courses.

Solve linear equations involving algebraic fractions and equations of more than 3 steps

- Solve [linear equations](#) involving more than 3 steps
- Solve [equations](#) that involve 2 or more [fractions](#)

Rearrange literal equations

- Change the subject of a formula

Solve quadratic equations using a variety of methods

- Solve equations of the form $ax^2 + bx + c = 0$ by factorisation and by completing the square
- Apply the quadratic formula $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$ to solve quadratic equations
- Apply the most appropriate method to solve a variety of [quadratic equations](#)
- Use substitution to verify solutions to quadratic equations
- Identify whether a given quadratic equation has real solutions, and if there are real solutions, whether or not they are equal
- Solve quadratic equations resulting from substitution into formulas in various contexts
- Model and solve word problems using quadratic equations in various contexts
- Solve equations that are reducible to quadratics

Solve linear simultaneous equations, both algebraically and graphically

- Solve linear [simultaneous equations](#) by finding the point of intersection of their [graphs](#)
- Solve linear simultaneous equations using algebraic techniques including substitution and elimination methods
- Model and solve word problems using simultaneous equations and interpret their solutions
- Describe an identity as an equation that is true for all values of the [pronumeral](#) and relate the identity to coincident [lines](#)
- Describe a contradiction as an equation that has no solutions and relate the contradiction to [parallel](#) lines

Linear relationships C (Path)

Outcomes

A student:

- develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly **MAO-WM-01**
- describes and applies transformations, the midpoint, gradient/slope and distance formulas, and equations of lines to solve problems (*Path: Adv*) **MA5-LIN-P-01**

Content

Stn (Standard), Adv (Advanced) and Ext (Extension) have been used to suggest paths for related Stage 6 courses.

Apply formulas to find the midpoint and gradient/slope of an interval on the Cartesian plane

- Apply the formula to find the midpoint of the interval joining 2 points on the Cartesian plane:

$$M(x,y) = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

- Use the relationship $m = \frac{\text{rise}}{\text{run}}$ to establish the formula for the gradient/slope (m) of the interval joining the 2 points (x_1, y_1) and (x_2, y_2) on the Cartesian plane: $m = \frac{y_2 - y_1}{x_2 - x_1}$
- Apply the [gradient](#) formula to find the gradient of the [interval](#) joining 2 points on the [Cartesian plane](#)

Apply the distance formula to find the distance between 2 points located on the Cartesian plane

- Apply knowledge of Pythagoras' theorem to establish the formula for the distance (d) between the 2 points (x_1, y_1) and (x_2, y_2) on the Cartesian plane: $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$
- Apply the distance formula to find the distance between 2 points on the Cartesian plane

Use various forms of the equation of a straight line

- Rearrange linear equations from gradient–intercept form ($y = mx + c$) to general form ($ax + by + c = 0$) and vice versa
- Find the x- and y-intercepts of a straight line in any form
- Graph the [equation](#) of a straight [line](#) in any form
- Use the point–gradient form ($y - y_1 = m(x - x_1)$) or the gradient–intercept form ($y = mx + c$) to find the equation of a line passing through a point (x_1, y_1) , with a given gradient m
- Use the gradient and the point–gradient form to find the equation of a line passing through 2 points
- Find the equation of a line that is [parallel](#) or [perpendicular](#) to a given line in any form
- Determine and justify whether 2 given lines are parallel or perpendicular

Solve problems by applying coordinate geometry formulas

- Solve problems including those involving geometrical figures by applying coordinate geometry formulas

Identify line and rotational symmetries

- Identify lines (axes) and rotational symmetry in plane [shapes](#)

- Identify line and rotational symmetry in various linear and [non-linear](#) graphs

Describe translations, reflections in an axis, and rotations through multiples of 90 degrees on the Cartesian plane, using coordinates

- Apply the notation P' to name the image resulting from applying a transformation to a point P on the Cartesian plane
- Determine and plot the coordinates for P' resulting from translating P one or more times
- Determine and plot the coordinates for P' resulting from reflecting P in either the x - or y -axis
- Determine and plot the coordinates for P' resulting from rotating P by a multiple of 90° about the origin

Non-linear relationships C (Path)

Outcomes

A student:

- develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly **MAO-WM-01**
- interprets and compares non-linear relationships and their transformations, both algebraically and graphically (*Path: Adv*) **MA5-NLI-P-01**

Content

Stn (Standard), Adv (Advanced) and Ext (Extension) have been used to suggest paths for related Stage 6 courses.

Graph parabolas and describe their features and transformations

- Use graphing applications to compare parabolas of the form $y = kx^2$, $y = kx^2 + c$, $y = k(x - b)^2$ and $y = k(x - b)^2 + c$, and describe their features and transformations
- Find x - and y -intercepts algebraically, where appropriate, for the graph of $y = ax^2 + bx + c$, given a, b and c
- Determine the equation of the axis of symmetry of a parabola using either the formula $x = -\frac{b}{2a}$ or the midpoint of the x -intercepts
- Find the [coordinates](#) of a parabola's [vertex](#) using a variety of methods
- Graph quadratic relationships of the form $y = ax^2 + bx + c$ by identifying and applying features of parabolas and their equations without graphing software

Graph exponentials and describe their features and transformations

- Use graphing applications to graph exponential relationships of the form $y = k(a)^x + c$ and $y = k(a)^{-x} + c$ for integer values of k , a and c (where $a > 0$ and $a \neq 1$), and compare and describe any relevant features

Graph hyperbolas and describe their features and transformations

- Use graphing applications to graph, compare and describe hyperbolic relationships of the form $y = \frac{k}{x}$ for integer values of k
- Use graphing applications to graph and describe a variety of hyperbolas, including where the equation is given in the form $y = \frac{k}{x} + c$ or $y = \frac{k}{x - b}$ for integer values of k , b and c

Graph circles and describe their features and transformations

- Derive the equation of a circle $x^2 + y^2 = r^2$ with centre $(0, 0)$ and radius r using the distance formula
- Identify and describe equations that represent circles with centre at the origin and radius of the circle r
- Graph circles of the form $x^2 + y^2 = r^2$, where r is the radius of the circle using graphing applications
- Establish the equation of the circle with centre (a, b) and radius r , and graph equations of the form $(x - a)^2 + (y - b)^2 = r^2$

- Find the centre and radius of a circle with the equation in the form $x^2 + y^2 + ax + by + c = 0$ by completing the square

Distinguish between different types of graphs by examining their algebraic and graphical representations and solve problems

- Identify and describe features of different types of graphs based on their [equations](#)
- Identify a possible equation from a graph and verify using graphing applications
- Find points where a [line](#) intersects with a [parabola](#), [hyperbola](#) or [circle](#), both graphically and algebraically

Graph and compare polynomial curves and describe their features and transformations

- Use graphing applications to graph and compare features of cubic equations of the form $y = ax^3 + c$, where a and c are integers
- Use graphing applications to graph a variety of equations of the form $y = kx^n$, where n is an integer and $n \geq 2$, and describe the effect on the shape of the curve where n is an odd or an even number
- Use graphing applications to graph curves of the form $y = kx^n + c$ and $y = k(x - b)^n$ where n is an integer and $n \geq 2$, and describe the transformations from $y = kx^n$

Polynomials (Path)

Outcomes

A student:

- develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly **MAO-WM-01**
- defines, operates with and graphs polynomials and applies the factor and remainder theorems to solve problems (*Path: Adv, Ext*) **MA5-POL-P-01**

Content

Stn (Standard), Adv (Advanced) and Ext (Extension) have been used to suggest paths for related Stage 6 courses.

Define and operate with polynomials

- Recognise a polynomial expression $a_n x^n + a_{n-1} x^{n-1} + \dots + a_2 x^2 + a_1 x + a_0$ where $n = 0, 1, 2, \dots$ and $a_0, a_1, a_2, \dots, a_n$ are real numbers
- Describe [polynomials](#) using terms such as degree, leading term, [coefficient](#) and leading coefficient, constant term, [monic](#) and non-monic
- Define a monic polynomial as having a leading coefficient of one
- Apply the notation $P(x)$ for polynomials and $P(c)$ to indicate the value of $P(x)$ for $x = c$
- Add, subtract and multiply polynomials

Divide polynomials

- Identify the [dividend](#), divisor, [quotient](#) and [remainder](#) in numerical division
- Divide a polynomial by a linear polynomial to find the quotient and remainder
- Express a polynomial in the form $P(x) = D(x)Q(x) + R(x)$, where $D(x)$ is the divisor, $Q(x)$ is the quotient and $R(x)$ is the remainder

Apply the factor and remainder theorems to solve problems

- Verify the remainder theorem and use it to find factors of polynomials and solve related problems
- Develop and apply the [factor theorem](#) to [factorise](#) particular polynomials completely and solve related problems
- Apply the factor theorem and division to find the zeroes of a polynomial $P(x)$ and solve $P(x) = 0$ (degree ≤ 4)
- State the maximum number of zeroes a polynomial of degree n can have

Graph polynomials

- [Graph](#) polynomials in factored form
- Graph quadratic, [cubic](#) and quartic polynomials by factorising and finding the zeroes
- Relate the term zeroes to polynomial [functions](#) and roots to polynomial [equations](#)
- Use graphing applications to determine the effect of single, double and triple roots of a polynomial equation $P(x) = 0$ on the shape of the graph for $y = P(x)$
- Graph polynomials using the sign of the leading term and the multiplicity of roots for the equation $P(x) = 0$
- Use graphing applications to compare the graphs of $y = -P(x)$, $y = P(-x)$, $y = P(x) + c$ and $y = kP(x)$ to the graph of $y = P(x)$

Logarithms (Path)

Outcomes

A student:

- develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly **MAO-WM-01**
- establishes and applies the laws of logarithms to solve problems (*Path: Adv*) **MA5-LOG-P-01**

Content

Stn (Standard), Adv (Advanced) and Ext (Extension) have been used to suggest paths for related Stage 6 courses.

Examine logarithms both numerically and graphically

- Define the term *logarithm*: the logarithm of a number to any positive base a is the index to which a is raised to give this number
- Recognise equivalence where $y = a^x$ is equivalent to $x = \log_a y$ where $a > 0$ and $a \neq 1$
- Translate statements expressing a number in [index](#) form into [equivalent](#) statements expressing the [logarithm](#) of the number
- Use graphing applications to compare and contrast graphs for the functions $y = a^x$ and $y = \log_a x$
- Generalise that $y = \log_a x$ is an increasing function when $a > 1$ and decreasing when $0 < a < 1$

Establish and apply the laws of logarithms to solve problems

- Deduce laws of logarithms from laws of [indices](#)
- Establish and use a variety of logarithmic results
- Apply the laws of logarithms to evaluate and simplify [expressions](#)
- Solve simple [equations](#) that involve exponents or logarithms
- Examine [logarithmic scales](#) and explain their use in various contexts

Functions and other graphs (Path)

Outcomes

A student:

- develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly **MAO-WM-01**
- uses function notation to describe and graph functions of one variable and graphs inequalities in one and 2 variables (*Path: Adv*) **MA5-FNC-P-01**

Content

Stn (Standard), Adv (Advanced) and Ext (Extension) have been used to suggest paths for related Stage 6 courses.

Define relations and functions, and use function notation

- Define a relation over 2 [sets](#) as an association between the elements of one set to the elements of another set
- Notate relations as a set of ordered pairs (x,y) of real numbers
- Define a [function](#) as a relation over 2 sets where each element of the first set is associated with exactly one element in the second set
- Apply the [vertical line test](#) on a [graph](#) to decide whether it represents a function
- Use the notation $f(x)$ when expressing a function
- Use the notation $f(c)$ to determine the value of $f(x)$ when $x = c$

Find the domain and range of a function and graph functions

- Define the domain as the set of all allowable values of x
- Define the range as the set of possible y -values as x varies over the domain of the function
- Determine and describe the [domain](#) and [range](#) for a variety of functions
- Use graphing applications to graph and compare functions of the form $y = f(x)$, $y = f(x) + c$, $y = f(x - b)$, $y = kf(x)$ and $y = f(ax)$, and describe their transformations

Graph regions corresponding to linear inequalities in one and 2 variables

- Graph linear inequalities of the form $ax + by > c$, testing whether the points satisfy the given inequality and shading appropriate regions

Trigonometry C (Path)

Outcomes

A student:

- develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly **MAO-WM-01**
- applies Pythagoras' theorem and trigonometry to solve 3-dimensional problems and applies the sine, cosine and area rules to solve 2-dimensional problems, including bearings (*Path: Stn, Adv*) **MA5-TRG-P-01**

Content

Stn (Standard), Adv (Advanced) and Ext (Extension) have been used to suggest paths for related Stage 6 courses.

Solve 3-dimensional problems involving right-angled triangles

- Apply [Pythagoras' theorem](#) to solve problems involving the [lengths](#) of the [edges](#) and diagonals of [rectangular](#) prisms and other 3-dimensional [objects](#)
- Apply trigonometry to solve problems involving right-angled [triangles](#) in 3 dimensions, including using [bearings](#) and [angles of elevation](#) and [depression](#)

Apply the sine, cosine and area rules to any triangle and solve related problems

- Use graphing applications to verify the sine rule $\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$ and that the ratios of a side to the sine of the opposite angle is a constant
- Apply the sine rule in a given triangle ABC to find the value of an unknown side
- Apply the sine rule in a given triangle ABC to find the value of an unknown angle (ambiguous case excluded): $\frac{\sin A}{a} = \frac{\sin B}{b} = \frac{\sin C}{c}$
- Use graphing applications to verify the cosine rule $c^2 = a^2 + b^2 - 2ab \cos C$.
- Apply the cosine rule to find the unknown sides for a given triangle ABC .
- Rearrange the formula to deduce that $\cos C = \frac{a^2 + b^2 - c^2}{2ab}$ and use this to find an unknown angle
- Use graphing applications to verify the area rule $A = \frac{1}{2}ab \sin C$.
- Apply the formula $A = \frac{1}{2}ab \sin C$, where a and b are the sides that form angle C to find the area of a given triangle ABC
- Solve problems involving finding unknown [angles](#) or [sides](#) in triangles (excluding right-angled triangles) by selecting and applying the appropriate rule

Trigonometry D (Path)

Outcomes

A student:

- develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly **MAO-WM-01**
- establishes and applies the properties of trigonometric functions and finds solutions to trigonometric equations (*Path: Adv*) **MA5-TRG-P-02**

Content

Stn (Standard), Adv (Advanced) and Ext (Extension) have been used to suggest paths for related Stage 6 courses.

Use the unit circle to define trigonometric functions and represent them graphically

- Redefine the [sine](#) and [cosine ratios](#) in terms of the [unit circle](#)
- Verify that the [tangent ratio](#) can be expressed as a [ratio](#) of the sine and cosine ratios
- Use graphing applications to examine the sine, cosine and tangent ratios for (at least) $0^\circ \leq x \leq 360^\circ$, and graph the results
- Use graphing applications to examine graphs of the sine, cosine and tangent functions for angles of any [magnitude](#), including negative [angles](#)
- Use the unit circle or graphs of trigonometric functions to establish and apply the relationships $\sin A = \sin(180^\circ - A)$, $\cos A = -\cos(180^\circ - A)$, and $\tan A = -\tan(180^\circ - A)$ for obtuse angles when $0^\circ \leq A \leq 90^\circ$
- Establish and apply the relationship $m = \tan \theta$, where m is the gradient of the line and θ is the angle of inclination of a line with the x -axis on the Cartesian plane

Solve trigonometric equations using exact values and the relationships between supplementary and complementary angles

- Derive and apply the exact sine, cosine and tangent ratios for angles of 30° , 45° and 60°
- Verify and use the relationships between the sine and cosine ratios of complementary angles in right-angled triangles: $\sin A = \cos(90^\circ - A)$ and $\cos A = \sin(90^\circ - A)$
- Find the possible [acute](#) and/or [obtuse angles](#), given a [trigonometric ratio](#)
- Apply the sine rule and area rule to find angles involving the ambiguous case

Area and surface area B (Path)

Outcomes

A student:

- develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly **MAO-WM-01**
- applies knowledge of the surface area of right pyramids and cones, spheres and composite solids to solve problems (*Path: Stn, Adv*) **MA5-ARE-P-01**

Related Life Skills outcomes: MALS-ARE-01

Content

Stn (Standard), Adv (Advanced) and Ext (Extension) have been used to suggest paths for related Stage 6 courses.

Solve problems involving surface areas

- Identify the [perpendicular](#) heights and slant heights of right pyramids and right cones
- Apply [Pythagoras' theorem](#) to find the slant heights, [base lengths](#) and perpendicular heights of right pyramids and right cones
- Solve problems involving the [surface areas](#) of right pyramids and compare methods of solution
- Apply the formula to find the curved surface area of right cones: $A = \pi rl$, where r is the length of the radius and l is the slant height
- Apply the formula to find the surface area of spheres: $A = 4\pi r^2$, where r is the length of the radius
- Solve problems involving the surface area of solids in a variety of contexts including composite solids

Volume B (Path)

Outcomes

A student:

- develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly **MAO-WM-01**
- applies knowledge of the volume of right pyramids, cones and spheres to solve problems involving related composite solids (*Path: Stn, Adv*) **MA5-VOL-P-01**

Content

Stn (Standard), Adv (Advanced) and Ext (Extension) have been used to suggest paths for related Stage 6 courses.

Solve problems involving volumes

- Find the volume of right pyramids and right cones by using an appropriate formula: volume of pyramid or cone $V = \frac{1}{3}Ah$, where A is the base area and h is the perpendicular height
- Find the volume of spheres by using an appropriate formula: volume of sphere $V = \frac{4}{3}\pi r^3$, where r is the length of the radius
- Apply knowledge of right pyramids, right cones and hemispheres to solve problems involving composite solids
- Solve practical problems related to the [volumes](#) and capacities of solids including right pyramids, right cones and spheres

Properties of geometrical figures B (Path)

Outcomes

A student:

- develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly **MAO-WM-01**
- establishes conditions for congruent triangles and similar triangles and solves problems relating to properties of similar figures and plane shapes (*Path: Ext*) **MA5-GEO-P-01**

Content

Stn (Standard), Adv (Advanced) and Ext (Extension) have been used to suggest paths for related Stage 6 courses.

Identify and explain congruence

- Identify figures as [congruent](#) figures if [translations](#), [reflections](#) and rotations can move one figure exactly on top of another
- Match the [sides](#) and [angles](#) of 2 congruent [polygons](#)
- Indicate 2 polygons are congruent using the symbol ($\equiv$) and name the vertices in matching order
- Determine that having equal radii is the condition for 2 [circles](#) to be congruent

Develop and use the conditions for congruent triangles

- Establish the 4 congruence tests for [triangles](#) (SSS, SAS, AAS and RHS)
- Use the congruence tests to identify a pair of congruent triangles from a selection of 2 or more triangles

Develop and apply the minimum conditions for triangles to be similar

- Examine the minimum conditions needed and establish the 4 tests for 2 triangles to be similar
- Apply the minimum conditions needed and determine whether 2 triangles are similar using an appropriate test

Establish and apply properties of similar shapes and solids

- Establish for 2 similar figures with similarity ratio $a:b$ that their areas are in the ratio $a^2:b^2$ and their volumes are in the ratio $a^3:b^3$
- Solve problems involving areas and [volumes](#) of similar [shapes](#) and solids

Apply logical reasoning to numerical problems involving plane shapes

- Apply geometrical facts, properties and relationships to find the sizes of unknown sides and angles of plane shapes in diagrams, providing appropriate reasons
- Define the exterior angle of a convex polygon
- Establish the sum of the exterior angles of any convex polygon is 360° and verify this result
- Apply the result for the sum of the [interior angles](#) of a triangle to find, by dissection, the sum of the interior angles of a polygon with more than 3 sides
- Apply the results for the interior and exterior angles of polygons to solve problems involving polygons

Properties of geometrical figures C (Path)

Outcomes

A student:

- develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly **MAO-WM-01**
- constructs proofs involving congruent triangles and similar triangles and proves properties of plane shapes (*Path: Ext*) **MA5-GEO-P-02**

Content

Stn (Standard), Adv (Advanced) and Ext (Extension) have been used to suggest paths for related Stage 6 courses.

Construct formal proofs involving congruent and similar triangles

- Construct formal proofs of the [congruence](#) of [triangles](#), preserving the matching order of vertices and drawing relevant conclusions
- Construct formal proofs of the [similarity](#) of triangles, preserving the matching order of vertices and drawing relevant conclusions

Apply logical reasoning to proofs involving plane shapes

- Recognise that a definition is the minimum amount of information needed to identify a particular figure
- Prove the properties of isosceles and equilateral triangles and special quadrilaterals from the formal definition of the [shapes](#)
- Prove and apply theorems and properties related to triangles and quadrilaterals
- Prove and apply tests for quadrilaterals
- Solve numerical and non-numerical problems in Euclidean geometry based on known assumptions and proven theorems

Circle geometry (Path)

Outcomes

A student:

- develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly **MAO-WM-01**
- applies deductive reasoning to prove circle theorems and solve related problems (*Path: Ext*) **MA5-CIR-P-01**

Content

Stn (Standard), Adv (Advanced) and Ext (Extension) have been used to suggest paths for related Stage 6 courses.

Prove and apply angle and chord properties of circles

- Apply terminology associated with [angles](#) in [circles](#)
- Identify the [arc](#) on which an angle at the centre or [circumference](#) stands
- Demonstrate that at any point on a circle there is a unique [tangent](#) to the circle, and that this tangent is [perpendicular](#) to the [radius](#) at the point of contact
- Prove [chord](#) properties of circles
- Prove angle properties of circles
- Apply chord and angle properties of circles to find unknown angles and [lengths](#) in diagrams

Prove and apply tangent and secant properties of circles

- Prove tangent and [secant](#) properties of circles
- Apply tangent and secant properties of circles to find unknown angles and lengths in diagrams

Introduction to networks (Path)

Outcomes

A student:

- develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly **MAO-WM-01**
- solves problems involving the characteristics of graphs/networks, planar graphs and Eulerian trails and circuits (*Path: Stn*) **MA5-NET-P-01**

Content

Stn (Standard), Adv (Advanced) and Ext (Extension) have been used to suggest paths for related Stage 6 courses.

Examine and describe a graph/network

- Describe a [network](#) as a collection of [objects](#) (nodes or vertices) interconnected by [lines](#) ([edges](#)) that can represent systems in the real world
- Examine real-world applications of networks such as social networks, supply chain networks and communication infrastructure, and explore other applications of networks
- Explain that the terms [graph](#) and *network* are interchangeable
- Identify and define elements of a graph including [vertex](#), [edge](#) and [degree](#)
- Explain that a given graph can be drawn in different ways

Define a planar graph and apply Euler's formula for planar graphs

- Define a [planar graph](#) as any graph that can be drawn in the plane so that no 2 edges cross
- Define a non-planar graph as a graph that can never be drawn in the plane without some edges crossing
- Demonstrate that some graphs that have crossing edges are still planar if they can be redrawn so that no 2 edges cross
- Identify the number of [faces](#) in a planar graph
- Describe and apply Euler's formula for planar graphs: $v - e + f = 2$, where v = vertices, e = edges and f = faces

Explain the concept of Eulerian trails and circuits in the context of the Königsberg bridges problem

- Explain that a [connected](#) graph is a graph that is in one piece, so that any 2 vertices are connected by a [path](#)
- Define a [walk](#) on a graph to be a sequence of vertices and edges of a graph
- Explain the difference between [trail](#), [circuit](#), path and [cycle](#)
- Relate the definition of a trail to an Eulerian trail as a walk in which every edge in the graph is included exactly once
- Relate the definition of a circuit to an Eulerian circuit that is defined as an Eulerian trail that ends at its starting point
- Relate Euler's Seven Bridges of Königsberg network problem to the definition of an Eulerian trail or circuit

Data analysis C (Path)

Outcomes

A student:

- develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly **MAO-WM-01**
- plans, conducts and reviews a statistical inquiry into a question of interest (*Path: Stn, Adv*) **MA5-DAT-P-01**

Content

Stn (Standard), Adv (Advanced) and Ext (Extension) have been used to suggest paths for related Stage 6 courses.

Plan and conduct a statistical inquiry into a question of interest

- Design an aim and hypothesis based on a question of interest
- Apply ethical and efficient methods for gathering and organising data
- Produce a report containing multiple data visualisations that provide insights and information

Examine reports of studies in digital media and elsewhere for information on their planning and implementation

- Examine and evaluate the appropriateness of sampling methods and [sample](#) size in reports with statements about a population and how they can affect the results of a survey
- Critically review surveys, polls and media reports for [accuracy](#) and/or [bias](#)
- Examine the use of statistics and the associated [probabilities](#) in shaping decisions made by governments and companies

Probability B (Path)

Outcomes

A student:

- develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly **MAO-WM-01**
- solves problems involving Venn diagrams, 2-way tables and conditional probability (*Path: Adv*) **MA5-PRO-P-01**

Content

Stn (Standard), Adv (Advanced) and Ext (Extension) have been used to suggest paths for related Stage 6 courses.

Solve problems involving Venn diagrams and 2-way tables

- Represent and interpret [data](#) in [Venn diagrams](#) for [mutually exclusive](#) and non-mutually exclusive [events](#)
- Construct Venn diagrams to represent all possible [combinations](#) of 2 [attributes](#) from given or collected data
- Interpret data in [2-way tables](#) to represent relationships between attributes
- Construct 2-way tables to represent the relationships between attributes
- Convert between representations of the relationships between 2 attributes in Venn diagrams and 2-way tables
- Define a [set](#) as a collection of distinct [objects](#)
- Use Venn diagrams, set language and notation for events, including A (or A' or A^c) for the complement of an event A , $A \cap B$ for ' A and B ' (the intersection of events A and B) and $A \cup B$ for ' A or B ' (the union of events A and B) and recognise mutually exclusive events

Use the language, 'if ... then', 'given', 'of' and 'knowing that', to examine conditional statements and identify common mistakes in interpreting the language

- Calculate the [probabilities](#) of events where a condition restricts the [sample space](#)
- Describe the effect of a given condition on the sample space
- Identify [conditional](#) statements used in descriptions of [chance](#) situations
- Explain the validity of conditional statements when describing chance situations, referring to [dependent](#) and [independent events](#)
- Identify and explain common misconceptions relating to chance experiments

Describe mutually and non-mutually exclusive events using specific language and calculate related probabilities

- Explain the difference between mutually exclusive and non-mutually exclusive events
- Describe compound events using the terms *inclusive or* and *exclusive or*
- Describe non-mutually exclusive events using the terminology *and*, *inclusive or* and *exclusive or*
- Describe events using the terms *at least*, *at most*, *not* and *and*
- Calculate the probability of compound events using strategies including Venn diagrams and 2-way tables